CS11212 **- Fall** 2024

Data Structures & Introduction to **Algorithms**

Analysis of Algorithms

part 1: Counting Operations

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What is an Algorithm?

A sequence of steps to solve a problem.

Example. Sequential Search is an algorithm for searching for an element in an array, which goes through all the elements one-by-one.

Python

```python
def search(mylist, k):
    for e in mylist:
        if e == k:
            return True
    return False
```

C++

```cpp
bool search(int mylist[], int k, int n) {
    for (int i = 0; i < n; i++)
        if (mylist[i] == k)
            return true;
    return false;
}
```

Java

```java
public static boolean search(int[] mylist, int k) {
    for (int i = 0; i < mylist.length; i++)
        if (mylist[i] == k)
            return true;
    return false;
}
```

The same algorithm implemented in different languages

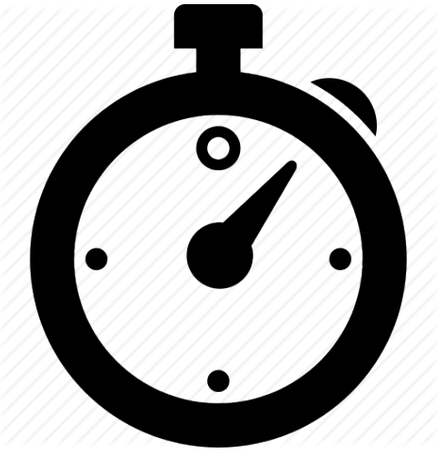
Comparing Algorithms

Q. Given two algorithms **A** and **B**, how do we know which is faster?

A. Implement and run both and compare the time each takes!

Experimental Analysis

To compare two algorithms, we can implement them, run them and compare their running times.



Challenges.

- The running time of a program is *hardware and software dependent*.
  We need to run both algorithms on the same machine (or on machines with the same specs), using the same programming language, the same compiler, etc.

- The running time of a program depends on the *input size* and on the *input type*.
  We need to run the programs as many times as needed to cover all possible input sizes and types that might affect the behavior of the programs.

- Running the programs *might take a long time*!
  Takes as long as the fastest of the two programs requires.

Which program runs faster?

Program A:

```
x = 1;
y = 2;
sum = x + y;
```

4 operations

Program B:

```
x = 1;
y = 2;
z = 3;
k = 4;
m = 5;
n = 6;
x = x + y;
x = x + z;
x = x + k;
x = x + m;
X = x + n;
```

16 operations

Theoretical Analysis

To compare two algorithms, *count* the number of operations each one performs.

Problem. Sometimes it is very difficult to count the number of operations or come up with a model for that.

Solution. Perform experimental analysis!

How Many Operations?

1×1	`i = 0;`
1×1	`sum = 0;`
1×11	`while (i < 10) {`
2×10	`  sum += i;`
2×10	`  i += 1;`
	`}`

$2 + (1 \times 11) + (4 \times 10) =$
53 operations

1×1	`i = 0;`
1×1	`sum = 0;`
1×21	`while (i < 20) {`
2×20	`  sum += i;`
2×20	`  i += 1;`
	`}`

$2 + (1 \times 21) + (4 \times 20) =$
103 operations

For simplicity, we will say:

- the left code performed the `sum += i` operation 10 times.
- the right code performed the `sum += i` operation 20 times.

We will always pick a certain operation to be the basis for our cost model.

How Many Operations?

How many times does `sum += i` get executed?

```
i = 0;
sum = 0;
while (i<5) {
   sum += i;
   i += 1;
}
```

5 times

```
i = 10;
sum = 0;
while (i>0) {
   sum += i;
   i -= 1;
}
```

10 times

```
i = 0;
sum = 0;
while (i<n) {
   sum += i;
   i += 1;
}
```

n times

Note: In all of the examples, *n* is assumed to be positive

How Many Operations?

How many times does `op()` get called?

```
i = 100;
while (i<n) {
   op();
   i += 1;
}
```

$n - 100$ times

for all $n > 100$ and 0 otherwise

```
i = 0;
while (i<n) {
   op();
   i += 5;
}
```

$\lceil n / 5 \rceil$ times

```
i = 100;
while (i<n) {
   op();
   i += 5;
}
```

$\lceil (n - 100) / 5 \rceil$ times

for all $n > 100$ and 0 otherwise

How Many Operations?

How many times does `op()` get called?

```
for (int i=0; i<n; i++)
   op();
```

n

```
for (int i=0; i<n; i+=5)
   op();
```

$\lceil n/5 \rceil$

How Many Operations?

How many times does `op()` get called?

```
for (int i=0; i<n; i++) {
   op();
   op();
}
```

$2n$

```
for (int i=0; i<n; i+=3) {
   op();
   op();
   op();
}
```

n

assuming n is a multiple of 3. If not, then the answer is: $\lceil n/3 \rceil \times 3$

How Many Operations?

How many times does `op()` get called?

```
for (int i=0; i<n; i++) {
   for (int j=0; j<n; j++)
      op();
}
```

n^2

```
for (int i=0; i<n; i++)
     op();
for (int j=0; j<n; j++)
     op();
```

$2n$

How Many Operations?

How many times does `op()` get called? (assuming n is a multiple of 2)

```
for (int i = 10; i < n; i++) {
   for (int j = 5; j < n; j += 2)
      op();
}
```

$$(n - 10) \times \frac{1}{2}(n - 5)$$

for all $n > 10$, 0 otherwise

How Many Operations?

How many times does `op()` get called? (assuming n is a multiple of 2)

```
for (int i = 0; i < n; i++) {
   for (int j = 0; j < n; j += 2)
      op();

   for (int j = 0; j < n; j += 2)
      op();
}
```

$$n \times (\frac{1}{2}n \ + \ \frac{1}{2}n) = n^2$$

If n is not a multiple of 2, the answer is: $n \times (\lceil \frac{1}{2}n \rceil \ + \ \lceil \frac{1}{2}n \rceil)$

How Many Operations?

How many times does `op()` get called? (assuming n is a multiple of 2)

```
for (int i = 0; i < n; i++)
  for (int j = 0; j < n; j += 2)
    for (int k = 10; k < n; k++)
      op();
```

$$n \times \frac{1}{2}n \times (n - 10) = \frac{1}{2}n^3 - 5n^2$$

for all $n > 10$, 0 otherwise

How Many Operations?

How many times does `op()` get called? (assuming *n* is a multiple of 2)

```
for (int i = 0; i < n*n; i++)
     op();

for (int i = 0; i < n; i += 2)
   for (int j = 0; j < n; j += 2)
      op();
```

$$n^2 + (\frac{1}{2}n \times \frac{1}{2}n)$$

$$= n^2 + \frac{1}{4}n^2$$

$$= \frac{5}{4}n^2$$

How Many Operations?

How many times does `op()` get called?

```
for (int i = 0; i < n; i++)
   for (int j = i; j < i + 7; j++)
      op();
```

$7n$

(the inner loop always repeats 7 times, regardless of what the value of *i* is)

```
for (int i = 0; i*i < n; i++)
     op();
```

$\sqrt{n}$

(the loop stops when $i^2 = n$ i.e. when $i = \sqrt{n}$)

How Many Operations?

How many times does `op()` get called? (assuming n is a power of 2)

```
for (int i = 1; i <= n; i *= 2)
    op();
```

$$i = 1, \quad 2, \quad 4, \quad 8, \quad \ldots, \quad \tfrac{1}{2}n, \quad n$$
$$= 2^0, \quad 2^1, \quad 2^2, \quad 2^3, \quad \ldots, \quad 2^{k-1}, \quad 2^k$$

These are $k + 1$ steps, where $2^k = n$ i.e. $k = \log_2(n)$
Total number of times `op()` is called = $\log_2(n) + 1$

```
for (int i = n; i >= 1; i /= 2)
    op();
```

$$i = n, \quad \tfrac{1}{2}n, \quad \tfrac{1}{4}n, \quad \ldots, 8, \quad 4, \quad 2, \quad 1$$
$$= 2^k, \quad 2^{k-1}, 2^{k-2}, \quad \ldots, 2^3, 2^2, 2^1, 2^0$$

These are $k + 1$ steps, where $2^k = n$ i.e. $k = \log_2(n)$
Total number of times `op()` is called = $\log_2(n) + 1$

How Many Operations?

How many times does `op()` get called? (assuming n is a power of 3)

```
for (int i = 1; i <= n; i *= 3)
    op();
```

$i =$	1,	3,	9,	27,	… ,	n
$=$	3^0,	3^1,	3^2,	3^3,	… ,	3^k

These are $k + 1$ steps, where $3^k = n$ i.e. $k = \log_3(n)$

Total number of times `op()` is called = $\log_3(n) + 1$

! In general:

```
for (i = 1; i <= whatever; i *= b)
    op();
```

$\lfloor \log_b(\text{whatever}) \rfloor + 1$

How Many Operations?

How many times does `op()` get called?

```
for (int i = 1; i <= n; i++)
   for (int j = 1; j <= i; j++)
      op();
```

X If the nested loops are *dependent*, we can't analyze each loop separately and then multiply them!

1 Trace

i	*number of* op() *calls*
1	1
2	2
3	3
...	...
n	n

Formulate a sum 2 Total $= 1 + 2 + 3 + \ldots + n$

$$= \sum_{i=0}^{n} i = \frac{n(n+1)}{2}$$

3 Solve the sum

Runtime Analysis Procedure

requires tracing skills
(structured programming?)

code → **trace** → **summation** → **answer**

requires math skills
(discrete mathematics?)

How Many Operations?

How many times does `op()` get called?

```
for (int i = 1; i <= n*n; i++)
   for (int j = 1; j <= i; j++)
      op();
```

`i`	*number of* `op()` *calls*
1	1
2	2
3	3
...	...
n∗n	n∗n

Total $= 1 + 2 + 3 + \ldots + n^2$

$$= \sum_{i=0}^{n^2} i = \frac{n^2(n^2+1)}{2}$$

! A very frequently encountered sum:

$$\sum_{i=0}^{\star} i = \frac{\star\,(\star+1)}{2}$$

How Many Operations?

How many times does `op()` get called?

```
for (int i = 1; i <= n; i++)
   for (int j = 1; j <= i; j++)
      for (int k = 1; k <= i; k++)
         op();
```

`i`	*number of* `op()` *calls*
1	1 x 1
2	2 x 2
3	3 x 3
...	...
n	n x n

Total $= 1^2 + 2^2 + 3^2 + \ldots + n^2$

$$= \sum_{i=0}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

see the math cheatsheet

How Many Operations?

How many times does `op()` get called?

```
for (int i = 1; i <= n; i *= 2)
  for (int j = 1; j <= i; j++)
    op();
```

`i`	*number of* `op()` *calls*
$1 = 2^0$	$1 = 2^0$
$2 = 2^1$	$2 = 2^1$
$4 = 2^2$	$4 = 2^2$
$8 = 2^3$	$8 = 2^3$
...	...
$n = 2^{\log_2 n}$	$n = 2^{\log_2 n}$

$$\text{Total} = 2^0 + 2^1 + 2^2 + \ldots + 2^{\log_2 n}$$

$$= \sum_{i=0}^{\log_2 n} 2^i = 2^{\log_2 n + 1} - 1 = 2^1 \times 2^{\log_2 n} - 1 = 2n - 1$$

see the math cheatsheet

How Many Operations?

```
bool foo(int n) {
   int random = rand() % 2;
   if (random == 0) {
       for (int i = 0; i < n; i++)
           op();
   } else
       op();
}
```

Best Case: `op()` is called 1 time (if random = 1)
Worst Case: `op()` is called n times (if random = 0).

Average Case: $P(0) \times cost(0) + P(1) \times cost(1)$

probability of random = 0 | # of op() calls if random = 0 | probability of random = 1 | # of op() calls if random = 1

$$\frac{1}{2} \times n + \frac{1}{2} \times 1 = \frac{1}{2}n + \frac{1}{2} = \frac{1}{2}(n+1)$$

How Many Operations?

```
bool search(int a[], int k, int n) {

   for (int i = 0; i < n; i++)
      if (a[i] == k)
         return true;
   return false;
}
```

Let's consider *comparisons with k* as the basis for our analysis.

Best Case: 1 comparison (k is the first element in the list).
Worst Case: n comparisons (k is not in the list).

Average Case: $\sum_{i=0}^{n-1} \mathrm{P}(i) \times \mathrm{cost}(i)$

probability of finding k at index i

number of operations if k is found at index i

Assuming k is equally likely to appear at any index:

$$= (\frac{1}{n} \times 1) + (\frac{1}{n} \times 2) + \ldots + (\frac{1}{n} \times n)$$

$$= \frac{1}{n} \times (\frac{n(n+1)}{2}) = \frac{1}{2}(n+1)$$

How Many Operations?

```
bool isSorted(int a[], int n) {

   for (int i = 1; i < n; i++)
      if (a[i - 1] > a[i])
         return false;
   return true;
}
```

Let's consider *comparisons between array elements* as the basis for our analysis.

Best Case: 1 comparison (first two elements are not in order).
Worst Case: n - 1 comparisons (list is in order).

Average Case: Not straightforward!

! We will focus on *best case* and *worst case* analysis in this course.

CS11212 **- Fall** 2023

Data Structures & Introduction to **Algorithms**

Analysis of Algorithms

part 2: Asymptotic Analysis

Ibrahim Albluwi

Which is better?

A

```
for (int i=0; i < 50 * n; i++)
   op();
```

$50n$

B

```
for (int i=0; i < n * n; i++)
   op();
```

n^2

We expressed the number of operations performed by each program as $T_A(n) = 50n$ and $T_B(n) = n^2$, which are two functions that have different values depending on the value of the input size n.

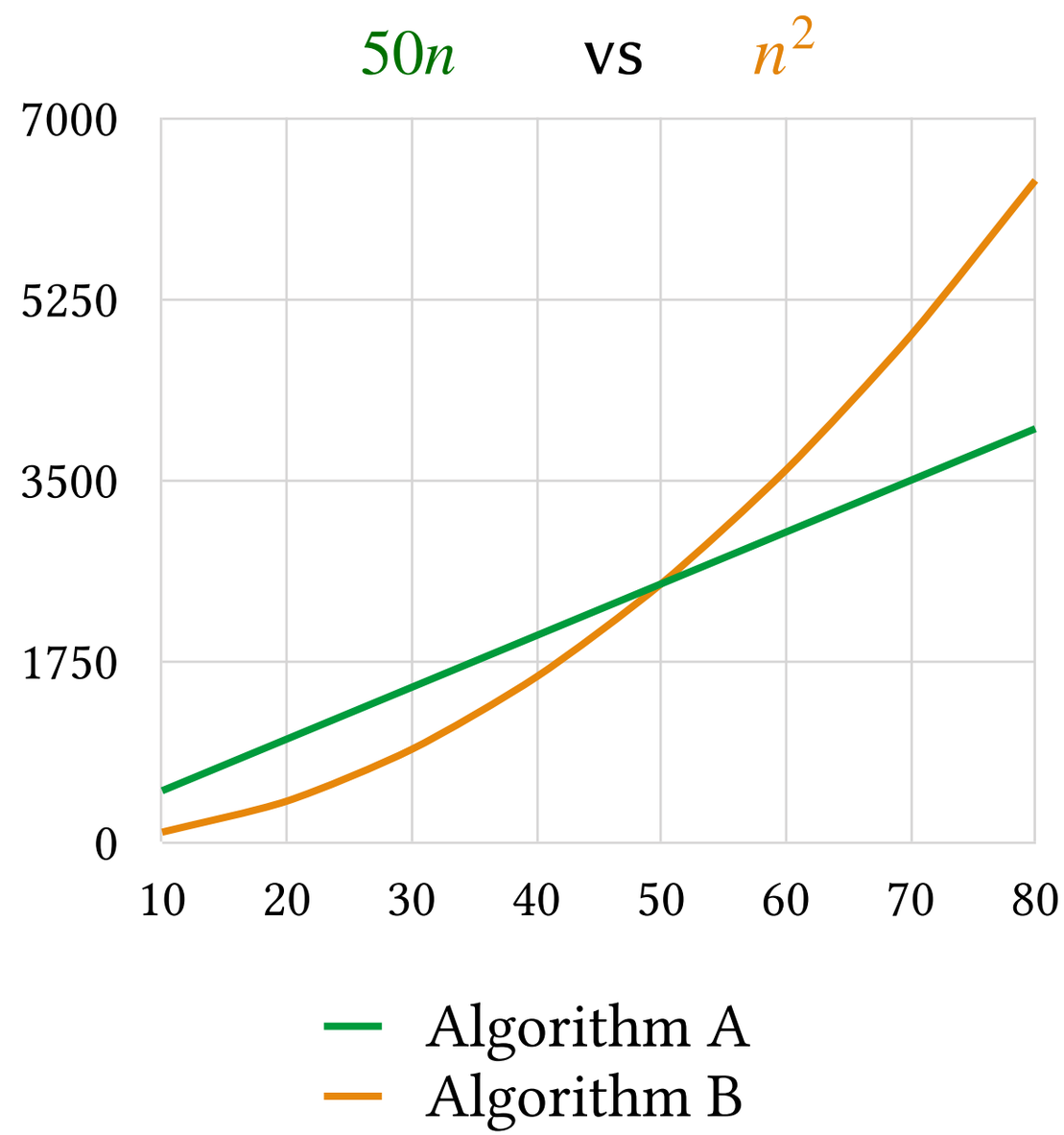
? Which function represents a better running time (less performed operations)?

Which is better?

	$50n$	n^2
n	Algorithm A	Algorithm B
10	500	100
20	1000	400
30	1500	900
40	2000	1600
50	2500	2500
60	3000	3600
70	3500	4900
80	4000	6400
90	4500	8100

$50n$ is worse

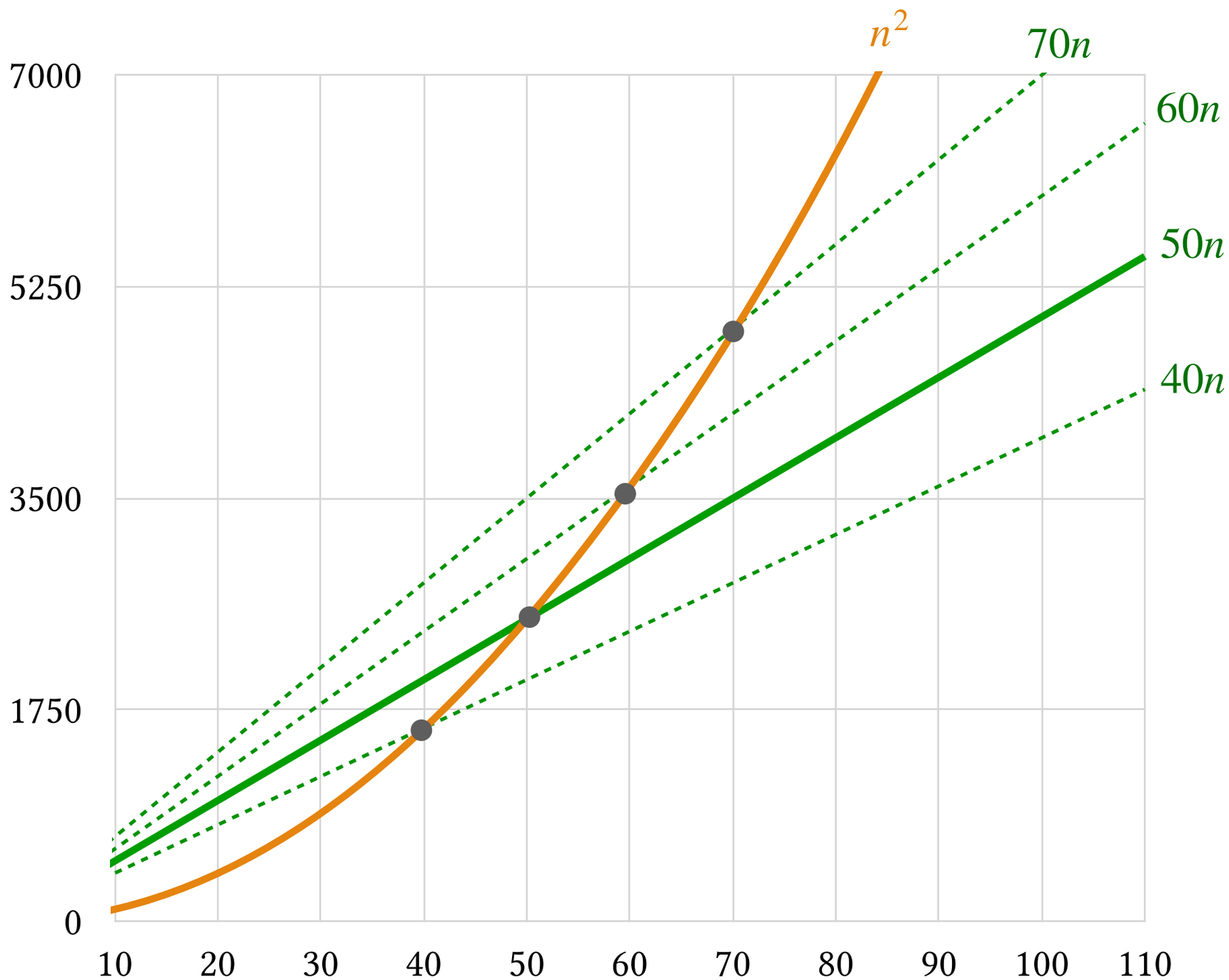
n^2 is worse



! n^2 grows faster than $50n$.

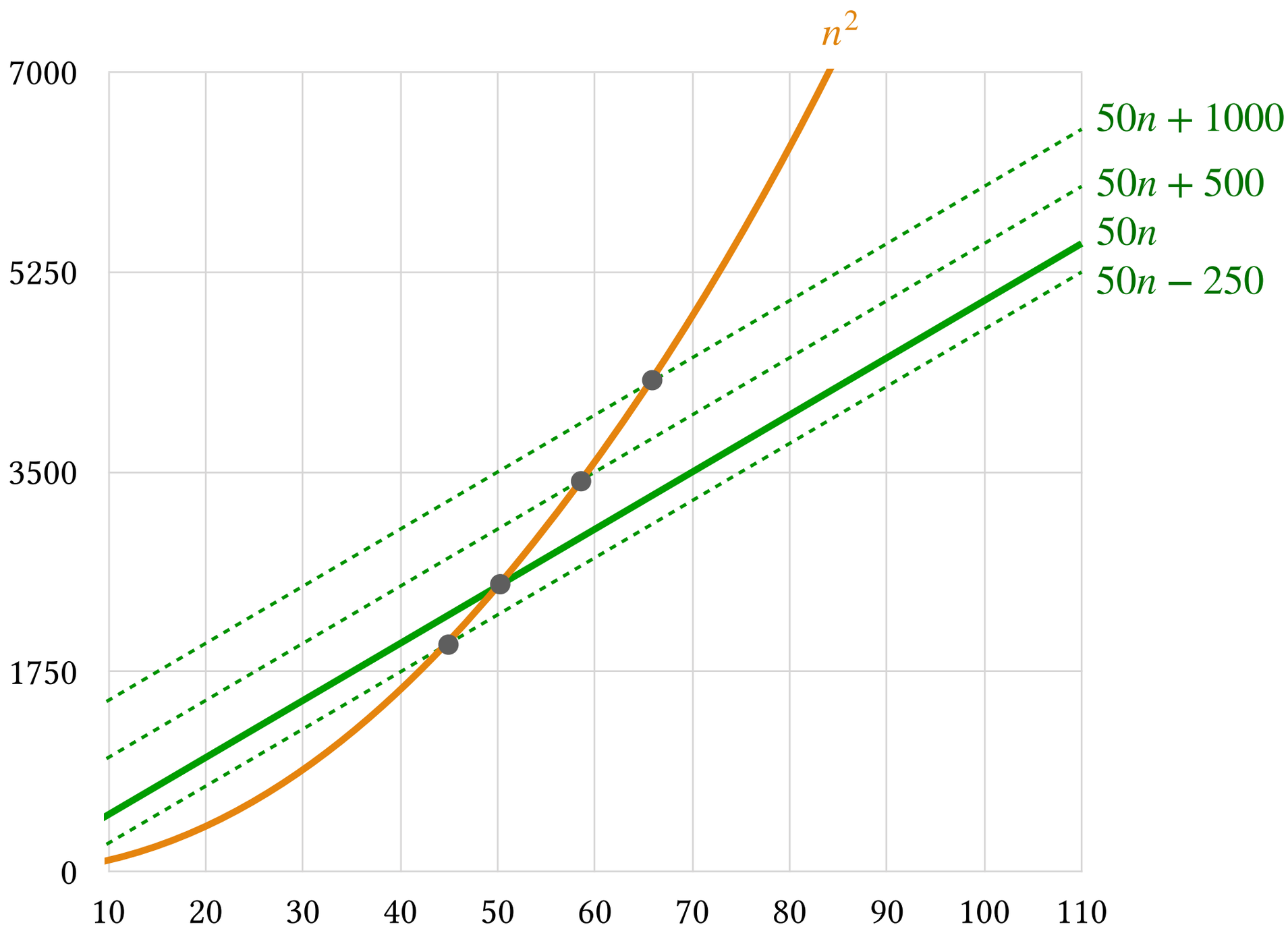
n^2 must at some point become worse (perform more operations) than $50n$ forever (when n > 50 in this case)

Orders of Growth



! n^2 will at some point exceed cn regardless of what the value of c is.

Orders of Growth



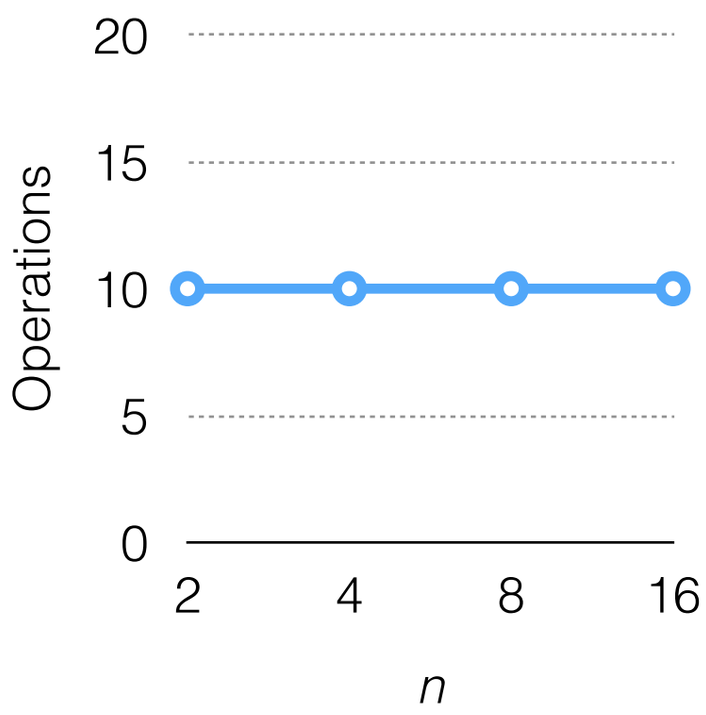
! n^2 will at some point exceed $cn + a$ regardless of what the values of c and a are.

Orders of Growth

Example. Assume $n \geq 10$ is the size of an array and we are interested in counting the number of array accesses an algorithm performs.

? How quickly does the number of operations performed grows when the input size grows (when the array size grows)?

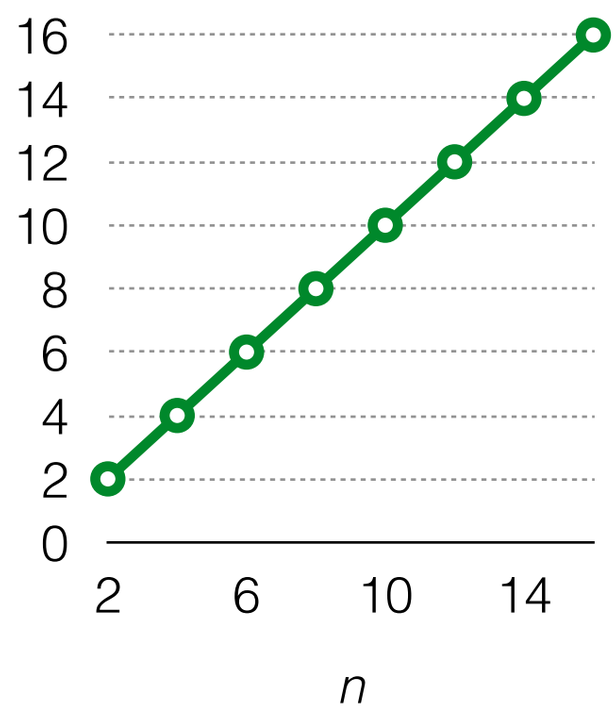
```
for (i=0; i<10; i++)
   sum += a[0];
```



No growth!

Always 10, regardless of the array size

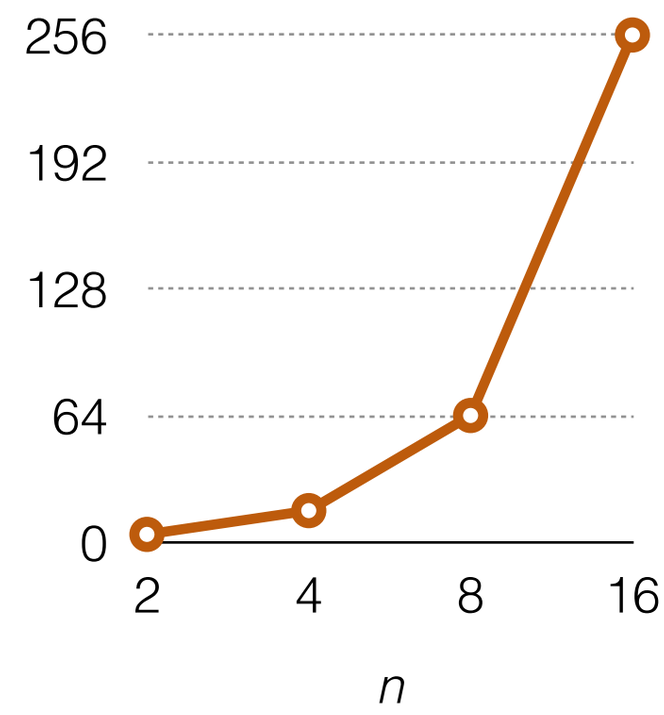
```
for (i=0; i<n; i++)
   sum += a[i];
```



Linear growth!

operations double when the array size doubles

```
for (i=0; i<n; i++)
  for (j=0; j<n; j++)
     sum += a[j];
```



Quadratic growth!

operations quadruple when the array size doubles

Lesson # 1

Look at the running time ***growth rate***!

! Classify algorithms based on the order of growth of their running time (ignoring the coefficients).

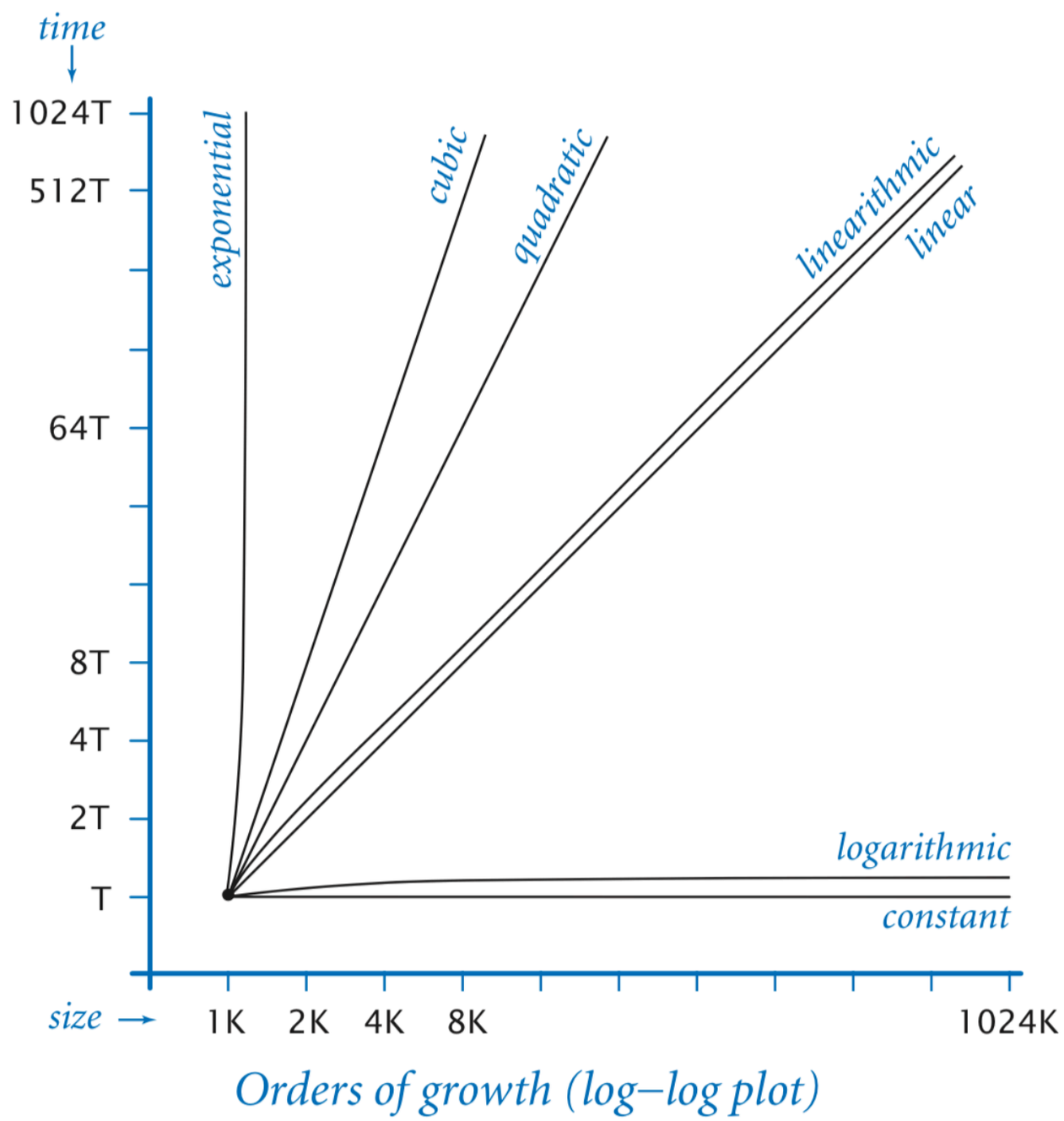
Example:

$5n^2,\ 30n^2,\ 7n^2,\ \text{etc}.$	have a *quadratic* order of growth.
$7n,\ 87n,\ 3n,\ \text{etc}.$	have a *linear* order of growth.
$3\log_2 n,\ 2\ln n,\ 10\log_{10} n,\ \text{etc}.$	have a *logarithmic* order of growth.

Examples of Growth Rates

graph by Kevin Wayne and Robert Sedgewick

	order of growth name	function
	constant	1
good	logarithmic	$\log(n)$
good		$\sqrt{n}$
fine	linear	n
fine	linearithmic	$n \log(n)$
fine		$n\sqrt{n}$
bad	quadratic	n^2
bad	cubic	n^3
horrible	exponential	2^n
horrible	exponential	3^n
horrible	factorial	$n!$



! constant < logarithmic < polynomial < exponential < factorial < n^n

$\log_b(n)$ $\quad$ n^c $(c > 0)$ $\quad$ c^n $(c > 1)$

Lower Order Terms? *Really?*

If the running time of an algorithm is given by:

$$n^2 \quad + \quad 100n \quad + \quad \log_{10}(n) \quad + \quad 1000$$

What is the *most* important term?

value of n	n^2	$100n$	$\log_{10}(n)$	1000
1	1	10^2	0	10^3
10	10^2	10^3	1	10^3
100	10^4	10^4	2	10^3
1000	10^6	10^5	3	10^3
100000	10^{10}	10^7	5	10^3

Assume
1 op requires
10^{-6} seconds:

$10^{10} \times 10^{-6}$ $=10^4$ seconds		$10^7 \times 10^{-6}$		5×10^{-6}		$10^3 \times 10^{-6}$
2.78 Hours	+	10 sec	+	0.000005 sec	+	0.001 sec

Lower Order Terms? *Really?*

If the running time of an algorithm is given by:

$$n^2 \quad + \quad 100n \quad + \quad \log_{10}(n) \quad + \quad 1000$$

What is the *most* important term?

value of n	n^2	$100n$	$\log_{10}(n)$	1000
1	1	10^2	0	10^3
10	10^2	10^3	1	10^3
100	10^4	10^4	2	10^3
1000	10^6	10^5	3	10^3
100000	10^{10}	10^7	5	10^3

n^2 dominates when the input size is large!

$10^{10} \times 10^{-6}$		$10^7 \times 10^{-6}$		5×10^{-6}		$10^3 \times 10^{-6}$
$=10^4$ seconds						
2.78 Hours	+	10 sec	+	0.000005 sec	+	0.001 sec

Lower Order Terms? *Really?*

Running time in seconds assuming each operation takes 10^{-6} seconds to execute.

order of growth \ input size	10	10^2	10^3	10^4	10^5
$\log_2(n)$	3.3×10^{-6}	6.6×10^{-6}	10^{-5}	1.3×10^{-5}	1.7×10^{-5}
$\sqrt{n}$	3.2×10^{-6}	10^{-5}	3.1×10^{-5}	10^{-4}	3.2×10^{-4}
n	10^{-5}	10^{-4}	0.001	0.01	0.1
n^2	10^{-4}	0.01	1 *sec*	1.67 *min*	2.78 *hr*
n^3	0.001	1 *sec*	16.7 *min*	11.6 *days*	31.7 *years*
2^n	0.001	4×10^{16} *years*	!!	!!	!!
$n!$	3.36 *sec*	3×10^{144} *years*	!!	!!	!!
n^n	2.78 *hr*	!!	!!	!!	!!

Asymptotic
Analysis

Lesson # 2

When working with large input sizes, consider only the ***highest order term***.

! Drop all lower order terms and coefficients and express the running time using Big-O notation:

Example: $T(n) = 5n^2 + 3n + 1 \longrightarrow O(n^2)$

Example: $T(n) = 3n^3 + n\log_2(n) \longrightarrow O(n^3)$

Lesson # 2

When working with large input sizes, consider only the ***highest order term***.

Technically (not for this course):
The running time of the algorithm (as a function of the input size n) is bounded above (after some point) by a constant multiplied by n^2.

Informally (in this course):
The running time (as a function of the input size n) has cn^2 as the highest order term ($c > 0$ is a constant).

! Drop all lower order terms and coefficients and express the running time using Big-O notation:

Example: $T(n) = 5n^2 + 3n + 1 \longrightarrow O(n^2)$

Example: $T(n) = 3n^3 + n\log_2(n) \longrightarrow O(n^3)$

What is the order of growth as a function of n?

```
for (int i = 0; i < 100; i += 5)          O(1)
   for (int j = 0; j < n; j++)            O(n)
      for (int k = 0; k < 2 * n; k++)     O(n)
         op();
```

$$O(1) \times O(n) \times O(n) = O(n^2)$$

```
for (int i = 0; i < 100; i += 5) {        O(1)
   for (int j = 1; j < n; j += 2)         O(n)
      op();
   for (int k = 0; k < 2 * n; k++)        O(n)
      op();
   op();
}
```

$$O(1) \times (O(n) + O(n) + O(1)) = O(n)$$

How Many Operations?

How many times does `op()` get called?

```
for (int i = 1; i <= n; i++)
   for (int j = 1; j <= i; j *= 2)
      op();
```

`i`	*number of* `op()` *calls*
1	$\log_2(1) + 1$
2	$\log_2(2) + 1$
3	$\log_2(3) + 1$
…	…
n	$\log_2(n) + 1$

Total $= \log_2(1) + \log_2(2) + \log_2(3) + \ \ldots \ + \log_2(n) + (n \times 1)$

$= \log_2(1 \times 2 \times 3 \times \ldots \times n) + (n \times 1) = \log_2(n!) + n$

$= O(n \log n)$ ← Using Stirling's Approximation (see the math cheatsheet)

Which function grows faster?

$T(n) = n^5$	same	$H(n) = n^5 + n^4 + n^3 + n^2 + n$
$T(n) = 2^n$	same	$H(n) = 2^{n+1}$
$T(n) = \log_2(n^2)$		$H(n) = (\log_2(n))^2$
$T(n) = \log_2(n)$	same	$H(n) = \log_{10}(n)$
$T(n) = \log_2(n)$		$H(n) = \sqrt{n} = n^{0.5}$
$T(n) = n\log_2(n)$		$H(n) = n\sqrt{n}$
$T(n) = 2^{\log_2(n)}$		$H(n) = 2^{\log_{10}(n)}$
$T(n) = \log_2(n)$		$H(n) = \log_2(\log_2(n))$

Do the math!

? The speed of a machine is 10^6 operations per second. Given an algorithm that performs $\sim n \lg n$ operations, how much time will this algorithm (roughly) require if we run it on an input size of $n = 1000$?

$$\text{speed} = \frac{\text{number of operations}}{\text{time}}$$

$$10^6 = \frac{1000 \times \lg(1000)}{\text{time}}$$

$$\text{time} = \frac{1000 \times \lg(1000)}{10^6} = \frac{\lg(1000)}{10^3} \approx 0.01 \text{ sec}$$

Do the math!

? An algorithm that performs $\sim \sqrt{n}$ operations takes 10^{-4} seconds to run with an input of size $n = 10^4$. How long is this algorithm expected to take if the input size is $n = 10^8$?

$$\text{speed} = \frac{\text{number of operations}}{\text{time}}$$

$$\text{speed} = \frac{\sqrt{10^4}}{10^{-4}} = 10^6 \text{ operations per second}$$

$$10^6 = \frac{\sqrt{10^8}}{\text{time}} \quad \longrightarrow \quad \text{time} = \frac{\sqrt{10^8}}{10^6} = 0.01 \text{ sec}$$

Do the math!

? An algorithm that performs $\sim n^2$ operations required 10 seconds to run on a machine that performs 10^7 operations per second. How much time is an algorithm that performs $\sim n^3$ operations expected to take when run on the same machine and the same input?

$$\text{speed} = \frac{\text{number of operations}}{\text{time}}$$

$10^7 = \dfrac{n^2}{10} \longrightarrow n^2 = 10 \times 10^7 \longrightarrow n = 10^4$	find the input size on which the n^2 algorithm took 10 seconds
$10^7 = \dfrac{n^3}{\text{time}} \longrightarrow \text{time} = \dfrac{(10^4)^3}{10^7} = 10^5 \text{ sec} \approx 27.8 \text{ hours}$	use the computed input size to find the time taken by the n^3 algorithm